STRINGS I FUNCIONS

JAVASCRIPT - 0612 - FRONTEND -BLOC 1

strings

```
5   var num = 444;
6   var text = "Un text";
7   var text2 = "Un altre text altre";
8
9   console.log('////////////////////////////////////');
10  console.log('///////////cadenes de text //////////');
11  console.log('////////////////////////////////////');
12
13  console.log(typeof(num));
14  console.log(typeof(num.toString()));
15
16  console.log(text.toUpperCase());
17  console.log(text.toLowerCase());
18  console.log(text.length);
19
20
21  var unArray = ["Hola", "Adeu"];
22  console.log(unArray.length);
23
24  console.log(text + "-" + text2);
25  console.log(text.concat("-" + text2));
26
27  // var text2 = "Un altre text altre";
28  //              0123456789
29
30  var cerca = text2.indexOf("altre");
31  console.log(cerca);
32
33  cerca = text2.indexOf("cap");
34  console.log("indexOf Cap:", cerca);
```

```
////////////////////////////////////
///////////cadenes de text //////////
////////////////////////////////////
number
string
UN TEXT
un text
7
2
Un text-Un altre text altre
Un text-Un altre text altre
3
indexOf Cap: -1
```

strings

```
36   cerca = text2.lastIndexOf("altre");
37   console.log("last index Of:", cerca);
38
39   cerca = text2.search("altre");
40   console.log('search:', cerca);
41
42   cerca = text2.match("altre");
43   console.log('match:', cerca);
44
45   cerca = text2.match(/altre/g);
46   console.log(cerca);
47
48   cerca = text2.substr(1, 6);
49   console.log('substr:', cerca);
50
51   cerca = text2.charAt(1);
52   console.log('charAt:', cerca);
53
54   cerca = text2.charAt(6);
55   console.log(cerca);
56
57   cerca = text2.startsWith("Un");
58   console.log('startsWith:', cerca);
59
60   cerca = text2.endsWith("altre");
61   console.log('endsWith:', cerca);
```

```
last index Of: 14
search: 3
match: ▼ ['altre', index: 3, input: 'Un altre text altre', groups: undefined]
          0: "altre"
          groups: undefined
          index: 3
          input: "Un altre text altre"
          length: 1
        ▶ [[Prototype]]: Array(0)
▶ (2) ['altre', 'altre']
substr: n altr
charAt: n
r
startsWith: true
endsWith: true
```

https://developer.mozilla.org/es/docs/Web/JavaScript/Reference/Global_Objects/String/substr

strings

https://www.freecodecamp.org/espanol/news/el-split-de-javascript-como-dividir-una-cadena-de-caracteres-en-un-arreglo-con-js/

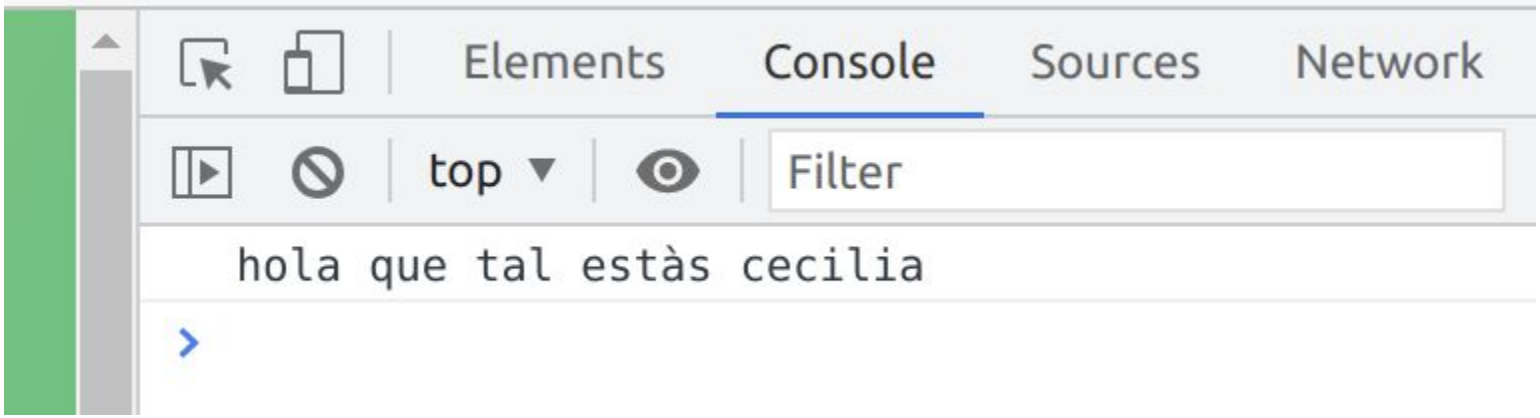
```
63  cerca = text2.includes("Altre");
64  console.log('includes:', cerca);
65
66  cerca = text2.replace("altre", "diferent");
67  console.log(cerca);
68  console.log(text2);
69
70  cerca = text2.split();
71  console.log(cerca);
72
73  cerca = text2.split(" ");
74  console.log(cerca);
75
76  text2 = "          Un altre text altre
77  cerca = text2.trim();
78  console.log(cerca);
```

```
includes: false
Un diferent text altre
Un altre text altre
▶ ['Un altre text altre']
▶ (4) ['Un', 'altre', 'text', 'altre']
Un altre text altre
>
```

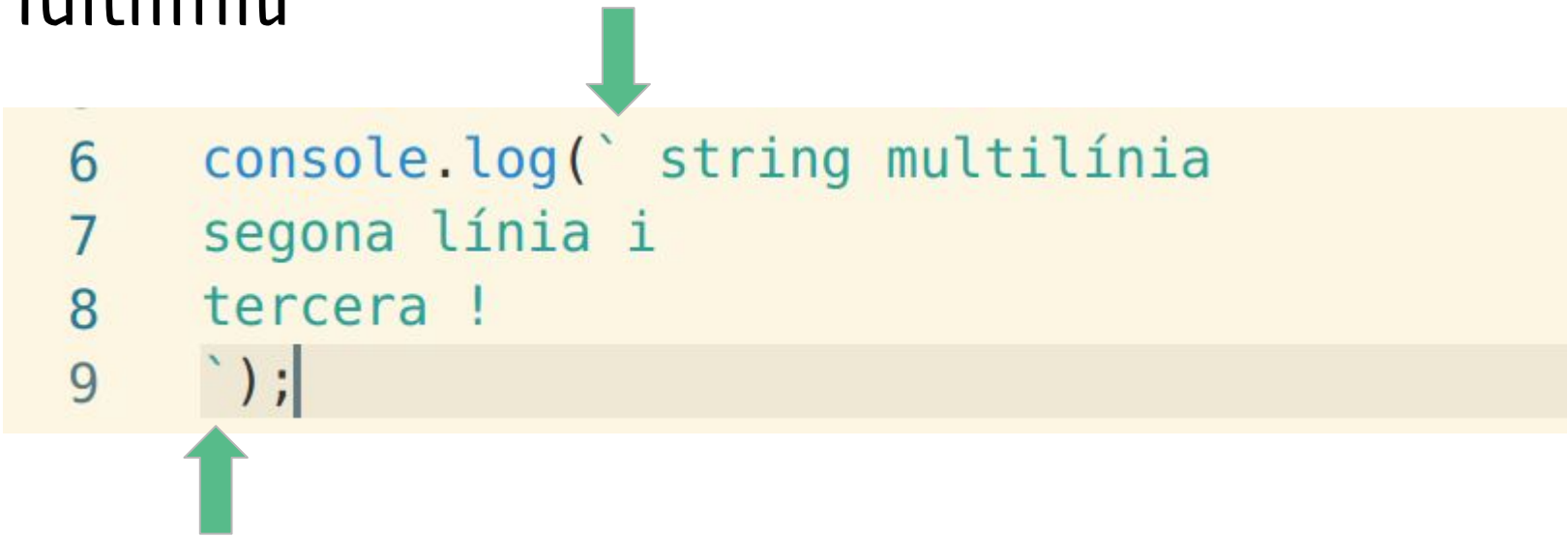
Plantilles amb strings (Backticks)

02-Fundamentals-Part-2 > starter > JS script.js > ...

```
1   let nom="cecilia";
2   let plantilla= `hola que tal estàs ${nom}`;
3   console.log(plantilla);
```



Multilínia




```
 string multilínia
segona línia i
tercera !
```

Un altre exemple

```
8   var nom=prompt("El teu nom?");
9   var cognom=prompt("El teu cognom?");
10
11  document.write(nom+"-"+cognom);
12
13  var plantilla = `
14      <h1> Una Plantilla </h1>
15      <h3> El nom : ${nom}</h3>
16      <h3> El cognom : ${cognom}</h3>
17
18  `;
19
20  document.write(plantilla);
```



Pilar-Mote

Una Plantilla

El nom : Pilar

El cognom : Mote

Number()

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to convert variables to numbers.
</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  var x1 = true;
  var x2 = false;
  var x3 = new Date();
  var x4 = "999";
  var x5 = "999 888";

  var n =
  Number(x1) + "<br>" +
  Number(x2) + "<br>" +
  Number(x3) + "<br>" +
  Number(x4) + "<br>" +
  Number(x5);
  document.getElementById("demo").innerHTML = n;
}
</script>

</body>
</html>
```

Click the button to convert variables to numbers.

Try it

1
0
1632821400146
999
NaN

parseFloat()

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to parse different strings.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  var a = parseFloat("10")
  var b = parseFloat("10.00")
  var c = parseFloat("10.33")
  var d = parseFloat("34 45 66")
  var e = parseFloat("   60   ")
  var f = parseFloat("40 years")
  var g = parseFloat("He was 40")

  document.getElementById("demo").innerHTML =
  a + "<br>" +
  b + "<br>" +
  c + "<br>" +
  d + "<br>" +
  e + "<br>" +
  f + "<br>" +
  g;
}
</script>

</body>
</html>
```

Click the button to parse

Try it

10
10
10.33
34
60
40
NaN

parseInt()

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to parse different strings.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  var a = parseInt("10") + "<br>";
  var b = parseInt("10.00") + "<br>";
  var c = parseInt("10.33") + "<br>";
  var d = parseInt("34 45 66") + "<br>";
  var e = parseInt("   60   ") + "<br>";
  var f = parseInt("40 years") + "<br>";
  var g = parseInt("He was 40") + "<br>";

  var h = parseInt("10", 10)+ "<br>";
  var i = parseInt("010")+ "<br>";
  var j = parseInt("10", 8)+ "<br>";
  var k = parseInt("0x10")+ "<br>";
  var l = parseInt("10", 16)+ "<br>";

  var n = a + b + c + d + e + f + g + "<br>" + h + i
+ j + k +l;
  document.getElementById("demo").innerHTML = n;
}
</script>

</body>
</html>
```

Click the button to parse different strings.

Try it

10
10
10
34
60
40
NaN

10
10
8
16
16

String()

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to convert different objects to
strings.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  var x1 = Boolean(0);
  var x2 = Boolean(1);
  var x3 = new Date();
  var x4 = "12345";
  var x5 = 12345;

  var res =
  String(x1) + "<br>" +
  String(x2) + "<br>" +
  String(x3) + "<br>" +
  String(x4) + "<br>" +
  String(x5);
  document.getElementById("demo").innerHTML = res;
}
</script>

</body>
</html>
```



Click the button to convert different objects to strings.

Try it

false
true
Tue Sep 28 2021 11:32:32 GMT+0200 (Hora d'estiu del Centre d'Europa)
12345
12345

isNaN()

```
<!DOCTYPE html>
<html>
<body>

<p>The isNaN() function returns true if the value is
NaN (Not-a-Number), and false if not.</p>

<p>Click the button to check whether a number is an
illegal number.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  var res = "";
  res = res + isNaN(123) + ": 123<br>";
  res = res + isNaN(-1.23) + ": -1.23<br>";
  res = res + isNaN(5-2) + ": 5-2<br>";
  res = res + isNaN(0) + ": 0<br>";
  res = res + isNaN('123') + ": '123'<br>";
  res = res + isNaN('Hello') + ": 'Hello'<br>";
  res = res + isNaN('2005/12/12') + ":
'2005/12/12'<br>";
  res = res + isNaN('') + ": ''<br>";
  res = res + isNaN(true) + ": true<br>";
  res = res + isNaN(undefined) + ": undefined<br>";
  res = res + isNaN('NaN') + ": 'NaN'<br>";
  res = res + isNaN(NaN) + ": NaN<br>";
  res = res + isNaN(0 / 0) + ": 0 / 0<br>";
  res = res + isNaN(null) + ": null<br>";

  document.getElementById("demo").innerHTML = res;
}
</script>

</body>
</html>
```

The isNaN() function returns true if the value is NaN (Not-a-Number), and false if not.

Click the button to check whether a number is an illegal number.

Try it

false: 123
false: -1.23
false: 5-2
false: 0
false: '123'
true: 'Hello'
true: '2005/12/12'
false: ''
false: true
true: undefined
true: 'NaN'
true: NaN
true: 0 / 0
false: null

Funcions

Example

```
function myFunction(p1, p2) {
  return p1 * p2;   // The function returns the product of p1
and p2
}
```

Sintaxi

```
function name(parameter1, parameter2, parameter3) {
  // code to be executed
}
```

Utilitzar una funció com una variable

```
function toCelsius(fahrenheit) {
  return (5/9) * (fahrenheit-32);
}
```

```
let x = toCelsius(77);
let text = "The temperature is " + x + " Celsius";
```

```
let text = "The temperature is " + toCelsius(77) + " Celsius";
```

Paràmetres amb valors per defecte

```
47  function multiplicaOSuma(p1, p2, opcional = false) {
48      var resultat;
49      if (opcional) {
50          resultat = p1 * p2;
51      } else {
52          resultat = p1 + p2;
53      }
54      return resultat; // The function returns the product of p1 and p2
55  }
```

```
56  var res = multiplicaOSuma(4, 3); //
57  console.log(res);
58  res = multiplicaOSuma(4, 3, true);
59  console.log(res);
```

Paràmetres rest spread

```
61  //// paràmetres REST SPREAD
62
63  function llistatAlumnes(alumne1, alumne2, ...resta_dalumnes) {
64      // em fa un array amb la resta de paràmetres que li passem
65
66      console.log("Alumne 1:", alumne1);
67      console.log("Alumne 2:", alumne2);
68      console.log(resta_dalumnes);
69
70  }
```

```
72  llistatAlumnes("Joan", "Jordi", "Josep", "Joel", "John", "Jan", "Jeroni");
73
74  var llista = ["Joan", "Jordi", "Josep", "Joel", "John", "Jan", "Jeroni"];
75  llistatAlumnes(...llista);
```

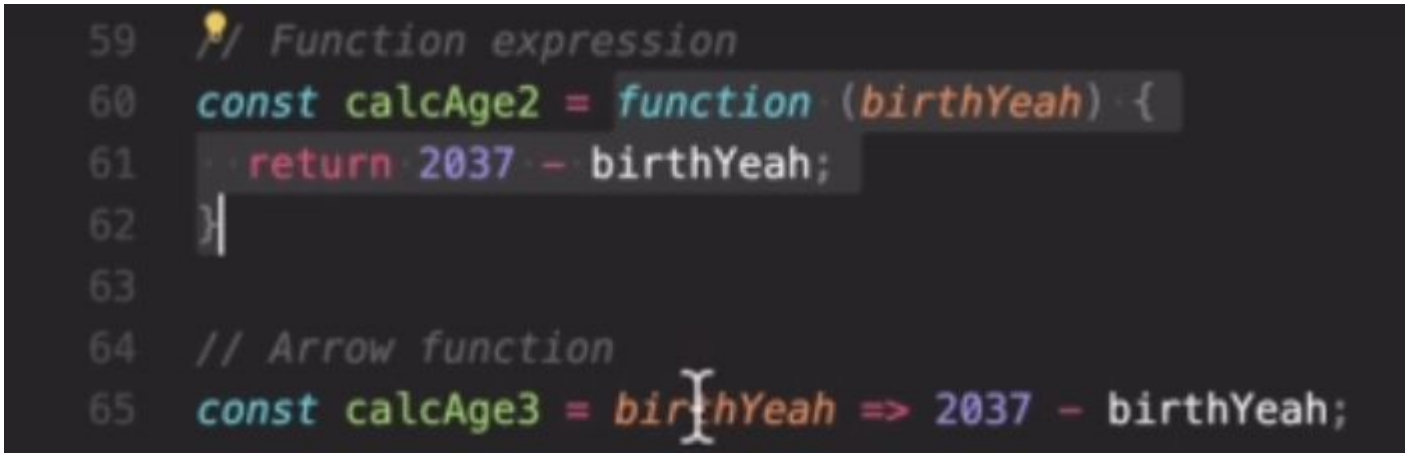
simplifiquem, funcions com expressió (anònimes)

diferència?? si és una expressió no cal declarar-la abans

```
42  function calcAge1(birthYeah) {
43    return 2037 - birthYeah;
44  }
```

```
48  // Function expression
49  const calcAge2 = function (birthYeah) {
50    return 2037 - birthYeah;
51  }
52  const age2 = calcAge2(1991);
53
```

Funcions fletxa



Funcions fletxa de més d'una línia

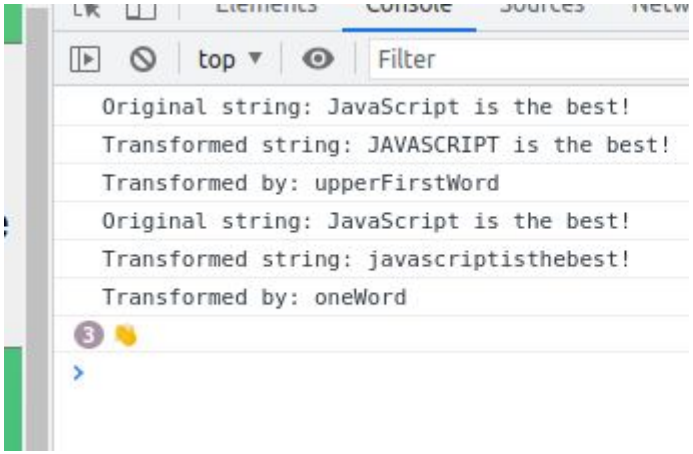
```
64  const yearsUntilRetirement = birthYeah => {
65    const age = 2037 - birthYeah;
66    const retirement = 65 - age;
67    return retirement;
68  }
```

Més d'un paràmetre en fletxa

```
64  const yearsUntilRetirement = (birthYeah, firstName)
    => {
65    const age = 2037 - birthYeah;
66    const retirement = 65 - age;
67    // return retirement;
68    return `${firstName} retires in ${retirement}
      years`;
69  }
```

Funcions que accepten funcions com a paràmetre

```
1   const oneWord = function (str) {
2     return str.replace(/ /g, '').toLowerCase();
3   };
4
5   const upperFirstWord = function (str) {
6     const [first, ...others] = str.split(' ');
7     return [first.toUpperCase(), ...others].join(' ');
8   };
9
10  // Higher-order function
11  const transformer = function (str, fn) {
12    console.log(`Original string: ${str}`);
13    console.log(`Transformed string: ${fn(str)}`);
14
15    console.log(`Transformed by: ${fn.name}`);
16  };
17
18  const transformer = function (str, fn) {
19    console.log(`Original string: ${str}`);
20    console.log(`Transformed string: ${fn(str)}`);
21
22    console.log(`Transformed by: ${fn.name}`);
23  };
24
25  transformer('JavaScript is the best!', upperFirstWord);
26  transformer('JavaScript is the best!', oneWord);
27
28  // JS uses callbacks all the time
29  const high5 = function () {
30    console.log('👋');
31  };
32  document.body.addEventListener('click', high5);
33  ['Jonas', 'Martha', 'Adam'].forEach(high5);
34
```



Funcions que retornen funcions

```
36  ////////////////////////////////////////
37  // Functions Returning Functions
38  const greet = function (greeting) {
39      return function (name) {
40        console.log(`${greeting} ${name}`);
41      };
42    };
43
44    const greeterHey = greet('Hey');
45    greeterHey('Jonas');
46    greeterHey('Steven');
47
48    greet('Hello')('Jonas');
49
50    // Challenge
51    const greetArr = greeting => name => console.log(`${greeting} ${name}`);
52
53    greetArr('Hi')('Jonas');
54
```

```
Hey Jonas
Hey Steven
Hello Jonas
Hi Jonas
>
```